on the pollutant (e.g., daily levels of $PM_{2.5}$) and the health response (e.g., daily mortality counts). There are several different functional forms, discussed below, that have been used to estimate C-R functions. The one most commonly used is the log-linear form, in which the natural logarithm of the health response is a linear function of the pollutant concentration.

For the purposes of estimating benefits, we are not interested in the C-R function itself, however, but the relationship between the change in concentration of the pollutant, Δx, and the corresponding change in the population health response, Δy. We want to know, for example, if the concentration of $PM_{2.5}$ is reduced by 10 μg/m³, how many premature deaths will be avoided? The relationship between Δx and Δy can be derived from the C-R function, as described below, and we refer to this relationship as a health impact function.

Many epidemiological studies, however, do not report the C-R function, but instead report some measure of the change in the population health response associated with a specific change in the pollutant concentration. The most common measure reported is the relative risk associated with a given change in the pollutant concentration. A general relationship between Δx and Δy can, however, be derived from the relative risk. The relative risk and similar measures reported in epidemiological studies are discussed in the sections below. The derivation of the relationship of interest for BenMAP - the relationship between Δx and Δy - is discussed in the subsequent sections.

C.2 Review Relative Risk and Odds Ratio

The terms relative risk and odds ratio are related but distinct. Table C-1 provides the basis for demonstrating their relationship.

Table C-1. Relative Risk and Odds Ratio Notation

Exposure	Fraction of Population		Adverse Effect Measure	
	Affected	Not Affected	Relative Risk	Odds
Baseline Pollutant Exposure	y_0	$1-y_0$	y_0/y_c	$y_0/(1-y_0)$
Control Pollutant Exposure	y_c	$1-y_c$		$y_c/(1-y_c)$

The "risk" that people with baseline pollutant exposure will be adversely affected (e.g., develop chronic bronchitis) is equal to y_0, while people with control pollutant exposure face a risk, y_0, of being adversely affected. The relative risk (RR) is simply:

$$RR = \frac{y_0}{y_c}$$

The odds that an individual facing high exposure will be adversely affected is:

$$Odds = \frac{y_0}{1 - y_0}$$

The odds ratio is then:

$$Odds\ Ratio = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)}$$

This can be rearranged as follows:

$$Odds\ Ratio = \frac{y_0}{y_c} \times \left(\frac{1-y_c}{1-y_0}\right) = RR \times \left(\frac{1-y_c}{1-y_0}\right)$$

As the risk associated with the specified change in pollutant exposure gets small (i.e., both y_0 and y_c approach zero), the ratio of $(1-y_c)$ to $(1-y_0)$ approaches one, and the odds ratio approaches the relative risk. This relationship can be used to calculate the pollutant coefficient in the C-R function from which the reported odds ratio or relative risk is derived, as described below.

C.3 Linear Model

A linear relationship between the rate of adverse health effects (incidence rate) and various explanatory variables is of the form:

$$y = \alpha + \beta \times PM$$

where α incorporates all the other independent variables in the regression (evaluated at their mean values, for example) times their respective coefficients. The relationship between the change in the rate of the adverse health effect from the baseline rate (y_0) to the rate after control (y_c) associated with a change from PM_0 to PM_c is then:

$$\Delta y = y_0 - y_c = \beta * (PM_0 - PM_c) = \beta * \Delta PM$$

For example, Ostro et al. (1991, Table 5) reported a $PM_{2.5}$ coefficient of 0.0006 (with a standard error of 0.0003) for a linear relationship between asthma and $PM_{2.5}$ exposure.

The lower and upper bound estimates for the $PM_{2.5}$ coefficient are calculated as follows:

$$\beta_{lowerbound} = \beta - (1.96 \times \sigma_\beta) = 0.0006 - (1.96 \times 0.0003) = 1.2 \times 10^{-5}$$

$$\beta_{upperbound} = \beta + (1.96 \times \sigma_\beta) = 0.0006 + (1.96 \times 0.0003) = 0.00119$$

It is then straightforward to calculate lower and upper bound estimates of the change in asthma.

C.4 Log-linear Model

The log-linear relationship defines the incidence rate (y) as:

$$y = B \times e^{\beta * PM}$$

Or, equivalently,

$$ln(y) = \alpha + \beta * PM,$$

where the parameter B is the incidence rate of y when the concentration of PM is zero, the parameter β is the coefficient of PM, ln(y) is the natural logarithm of y, and α = ln(B). Other covariates besides pollution clearly affect mortality. The parameter B might be thought of as containing these other covariates, for example, evaluated at their means. That is,

$$B = B_0 \times e^{\beta_1 x_1 + \cdots + \beta_n x_n}$$

where Bo is the incidence of y when all covariates in the model are zero, and x1, ... , xn are the other covariates evaluated at their mean values. The parameter B drops out of the model, however, when changes in y are calculated, and is therefore not important.

The relationship between ΔPM and Δy is:

$$\Delta y = y_0 - y_c = B\left(e^{\beta PM_0} - e^{\beta PM_c}\right)$$

This may be rewritten as:

$$\Delta y = B \times e^{\beta \bullet PM_0}\left(1 - e^{-\beta(PM_0 - PM_c)}\right) = y_0\left(1 - \frac{1}{\exp(\beta \times \Delta PM)}\right)$$

where y_0 is the baseline incidence rate of the health effect (i.e., the incidence rate before the change in PM).

The change in the incidence of adverse health effects can then be calculated by multiplying the change in the incidence rate, Δy, by the relevant population (e.g., if the rate is number per 100,000 population, then the relevant population is the number of 100,000s in the population).

When the PM coefficient (β) and its standard error (σβ) are published (e.g., Ostro et al., 1989), then the coefficient estimates associated with the lower and upper bound may be calculated easily as follows:

$$\beta_{lowerbound} = \beta - \left(1.96 \times \sigma_\beta\right)$$

$$\beta_{upperbound} = \beta + \left(1.96 \times \sigma_\beta\right),$$

Where the adjustment on the mean of ±1.96 times the standard error produces the 2.5^{th} and 97.5^{th} percentiles of the normal distribution, which are used to approximate a 95%

confidence interval. These values can be changed to capture different lower and upper bounds.

Epidemiological studies often report a relative risk for a given ΔPM, rather than the coefficient, β (e.g., Schwartz et al., 1995, Table 4). Recall that the relative risk (RR) is simply the ratio of two risks:

$$RR = \frac{y_0}{y_c} = e^{\beta \bullet \Delta PM}$$

Taking the natural log of both sides, the coefficient in the C-R function underlying the relative risk can be derived as:

$$\beta = \frac{In(RR)}{\Delta PM}$$

The coefficients associated with the lower and upper bounds (e.g., the 2.5th and 97.5th percentiles) can be calculated by using a published confidence interval for relative risk, and then calculating the associated coefficients.

Because of rounding of the published RR and its confidence interval, the standard error for the coefficient implied by the lower bound of the RR will not exactly equal that implied by the upper bound, so an average of the two estimates is used. The underlying standard error for the coefficient (σ_β) can be approximated by:

$$\sigma_{\beta,\, 2.5\ percentile} = \frac{\beta - \beta_{2.5\ percentile}}{1.96}$$

$$\sigma_{\beta,\, 97.5\ percentile} = \frac{\beta_{97.5\ percentile} - \beta}{1.96}$$

$$\sigma_\beta \cong \frac{\sigma_{\beta,\, 2.5\ percentile} + \sigma_{\beta,\, 97.5\ percentile}}{2}$$

C.5 Logistic Model

In some epidemiological studies, a logistic model is used to estimate the probability of an occurrence of an adverse health effect. Given a vector of explanatory variables, X, the logistic model assumes the probability of an occurrence is:

$$y = prob(occurrence \mid X \times \beta) = \left(\frac{e^{X \bullet \beta}}{1 + e^{X \bullet \beta}} \right),$$